

Society of St. Francis Xavier, Pilar's
FR. CONCEICAO RODRIGUES COLLEGE OF ENGINEERING

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SUPPLEMENTARY ANSWER BOOK



(To be filled by the Candidate)

College Seal & Date to be affixed within the square	
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(In Words)

Candidate's Seat No.

(In Figures)

Junior Supervisor's Name	Junior Supervisor's Full Signature with Date

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Question No.	Linear Discriminant Analysis (LDA):-
	LDA is a supervised Machine learning algorithm primarily used for dimensionality reduction and classification task.
	The goal is to find a linear combination of features that maximizes the separation between different classes while minimizing the spread (variance) within each class. Two criteria are used by LDA to create a new axis.
	Maximizing Mean difference
	Minimizing within class difference.

Step 1 Calculate the mean vector for each class

Step 2 Compute the covariance of each class

Step 3 Compute the within class scatter matrix

Step 4 Compute the between class scatter matrix

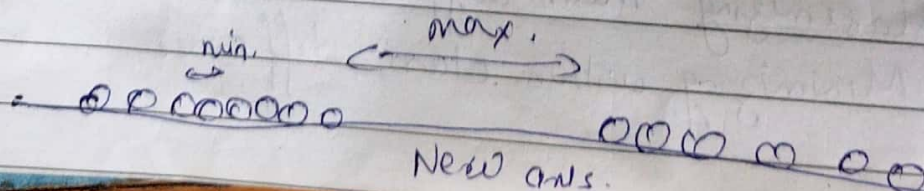
Step 5 Find eigen values and choose top eigen value.

Step 6 Find the eigen vectors

Step 7 Find the dot product of data points with eigen vector.

It is basically used to project the features in higher dimension space into a lower dimension space.

Here, LDA uses both the axes (X and Y) to create a new axis and projects data onto a new axis in a way to maximize the separation of the two categories and hence reducing the 2D graph into 1D graph



$$\Rightarrow 1 \quad \mu_1 = \frac{1}{N_1} \sum_{x \in w_1} x$$

$$\Rightarrow S_1 = \sum_{x \in w_1} (x - \mu_1)(x - \mu_1)^T / N - 1$$

$$\Rightarrow S_W = S_1 + S_2$$

$$\Rightarrow S_B = (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T$$

$$\Rightarrow S_W^{-1} S_B w = \lambda w \quad \text{find } \lambda \text{ for each class}$$

$$\Rightarrow \text{Sort the values of eigen values and select the top } k \text{ value}$$

$$(S_W^{-1} S_B - \lambda I) \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = 0$$

$$\Rightarrow \text{Dot product of Eigen vectors and original data}$$

ex → Compute the LD projection for the following two dimensional dataset

Samples for class w_1 :

$$X_1 = (x_1, x_2) = \{(4, 2), (2, 4), (2, 3), (3, 6), (4, 4)\}$$

Samples for class w_2 :

$$X_2 = (x_1, x_2) = \{(9, 10), (6, 8), (9, 5), (8, 7), (10, 8)\}$$

Solⁿ

Step 1

Mean of 1st class is

$$\mu_1 = \frac{1}{N_1} \sum_{x \in w_1} x = \frac{1}{5} \left[\begin{pmatrix} 4 \\ 2 \end{pmatrix} + \begin{pmatrix} 2 \\ 4 \end{pmatrix} + \begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 3 \\ 6 \end{pmatrix} + \begin{pmatrix} 4 \\ 4 \end{pmatrix} \right]$$

$$\mu_1 = \begin{pmatrix} 3 \\ 3.8 \end{pmatrix}$$

Mean of 2nd class is

$$\mu_2 = \frac{1}{N_2} \sum_{x \in w_2} x = \frac{1}{5} \left[\begin{pmatrix} 9 \\ 10 \end{pmatrix} + \begin{pmatrix} 6 \\ 8 \end{pmatrix} + \begin{pmatrix} 9 \\ 5 \end{pmatrix} + \begin{pmatrix} 8 \\ 7 \end{pmatrix} + \begin{pmatrix} 10 \\ 8 \end{pmatrix} \right]$$

$$\therefore \mu_2 = \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix}$$

Step 2

Calculate the Covariance matrix of 1st class

$$S_1 = \sum_{x \in w_1} \frac{(x - \mu_1)(x - \mu_1)^T}{N-1}$$

$$= \frac{1}{5-1} \left[\begin{pmatrix} 4 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \right] \left[\begin{pmatrix} 4 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \right]^T + \begin{pmatrix} 2 \\ 4 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \left[\begin{pmatrix} 2 \\ 4 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \right]^T + \begin{pmatrix} 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \left[\begin{pmatrix} 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \right]^T + \begin{pmatrix} 3 \\ 6 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \left[\begin{pmatrix} 3 \\ 6 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \right]^T + \begin{pmatrix} 4 \\ 4 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \left[\begin{pmatrix} 4 \\ 4 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \right]^T \right]$$

5-1

$$= \frac{1}{4} \left[\begin{pmatrix} 1 \\ -1.8 \end{pmatrix} \begin{pmatrix} 1 & -1.8 \end{pmatrix} + \begin{pmatrix} -1 \\ 0.2 \end{pmatrix} \begin{pmatrix} -1 & 0.2 \end{pmatrix} + \begin{pmatrix} -1 \\ -0.8 \end{pmatrix} \begin{pmatrix} -1 & -0.8 \end{pmatrix} + \begin{pmatrix} 0 \\ 2.2 \end{pmatrix} \begin{pmatrix} 0 & 2.2 \end{pmatrix} \right]$$

$$+ \begin{bmatrix} 1 & 0.2 \\ 0.2 & 0.04 \end{bmatrix} / 4 = \begin{bmatrix} 1 & -1.8 \\ -1.8 & 3.24 \end{bmatrix} + \begin{bmatrix} 1 & -0.2 \\ -0.2 & 0.04 \end{bmatrix} + \begin{bmatrix} 1 & 0.8 \\ 0.8 & 0.64 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 4.84 \end{bmatrix} + \begin{bmatrix} 1 & 0.2 \\ 0.2 & 0.04 \end{bmatrix} / 4 = \frac{1}{4} \begin{bmatrix} 4 & -1 \\ -1 & 8.8 \end{bmatrix}$$

$$S_1 = \begin{bmatrix} 1 & -0.25 \\ -0.25 & 2.2 \end{bmatrix} \quad \leftarrow \quad \frac{1}{4} \begin{bmatrix} 4 & -1 \\ -1 & 8.8 \end{bmatrix}$$

Similarly Calculate S_2

$$S_2 = \frac{\sum (x - \mu_2)(x - \mu_2)^T}{N-1}$$

$$\begin{aligned}
 S_2 = & \left[\begin{pmatrix} 9 \\ 10 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right] \left[\begin{pmatrix} 9 \\ 10 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right]^T + \\
 & \left[\begin{pmatrix} 6 \\ 8 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right] \left[\begin{pmatrix} 6 \\ 8 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right]^T + \\
 & \left[\begin{pmatrix} 9 \\ 5 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right] \left[\begin{pmatrix} 9 \\ 5 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right]^T + \\
 & \left[\begin{pmatrix} 8 \\ 7 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right] \left[\begin{pmatrix} 8 \\ 7 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right]^T \\
 & + \left[\begin{pmatrix} 10 \\ 8 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right] \left[\begin{pmatrix} 10 \\ 8 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right]^T
 \end{aligned}$$

4

$$\begin{aligned}
 S_2 = & \begin{bmatrix} 0.6 \\ 2.4 \end{bmatrix} \begin{bmatrix} 0.6 & 2.4 \end{bmatrix} + \begin{bmatrix} -2.4 \\ 0.4 \end{bmatrix} \begin{bmatrix} -2.4 & 0.4 \end{bmatrix} \\
 & + \begin{bmatrix} 0.6 \\ -2.6 \end{bmatrix} \begin{bmatrix} 0.6 & -2.6 \end{bmatrix} + \begin{bmatrix} -0.4 \\ -0.6 \end{bmatrix} \begin{bmatrix} -0.4 & -0.6 \end{bmatrix} \\
 & + \begin{bmatrix} 1.6 \\ 0.4 \end{bmatrix} \begin{bmatrix} 1.6 & 0.4 \end{bmatrix}
 \end{aligned}$$

4

$$\begin{aligned}
 S_2 = & \begin{bmatrix} 0.36 & 1.44 \\ 1.44 & 5.76 \end{bmatrix} + \begin{bmatrix} 5.76 & -0.96 \\ -0.96 & 0.16 \end{bmatrix} \\
 & + \begin{bmatrix} 0.36 & -1.38 \\ -1.38 & 5.29 \end{bmatrix} + \begin{bmatrix} 0.16 & 0.24 \\ 0.24 & 0.36 \end{bmatrix} + \begin{bmatrix} 2.56 & 0.64 \\ 0.64 & 0.16 \end{bmatrix}
 \end{aligned}$$

4

$$S_2 = \frac{1}{4} \begin{bmatrix} 9.2 & -0.02 \\ -0.02 & 13.72 \end{bmatrix}$$

$$S_2 = \begin{bmatrix} 2.3 & -0.005 \\ -0.005 & 3.38 \end{bmatrix}$$

with class scatter Matrix

$$S_w = S_1 + S_2 = \begin{bmatrix} 1 & -0.25 \\ -0.25 & 2.2 \end{bmatrix} + \begin{bmatrix} 2.3 & -0.05 \\ -0.05 & 3.3 \end{bmatrix}$$

$$= \begin{bmatrix} 3.3 & -0.3 \\ -0.3 & 5.5 \end{bmatrix}$$

Between class scatter Matrix

$$S_B = (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T$$

$$= \left[\begin{pmatrix} 3 \\ 3.8 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right] \left[\begin{pmatrix} 3 \\ 3.8 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right]^T$$

$$= \begin{bmatrix} -5.4 \\ -3.8 \end{bmatrix} \begin{bmatrix} -5.4 & -3.8 \end{bmatrix}$$

$$= \begin{bmatrix} 29.16 & 20.52 \\ 20.52 & 14.44 \end{bmatrix}$$

find Eigen values: $- S_w^{-1} S_B w = \lambda w$

$$\Rightarrow [S_w^{-1} S_B - \lambda I] = 0$$

$$\begin{bmatrix} 3.3 & -0.3 \\ -0.3 & 5.5 \end{bmatrix}^{-1} \begin{bmatrix} 29.16 & 20.52 \\ 20.52 & 14.44 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} 0.3045 & 0.0166 \\ 0.0166 & 0.1827 \end{bmatrix} \begin{bmatrix} 29.16 & 20.52 \\ 20.52 & 14.44 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$\begin{vmatrix} 9.2213 - \lambda & 6.489 \\ 4.2339 & 2.9794 - \lambda \end{vmatrix} = 0$$

$$(9.2213 - \lambda)(2.9794 - \lambda) - 6.489 \times 4.2339 = 0$$

$$\lambda^2 - 12.2007\lambda = 0 \Rightarrow \lambda(\lambda - 12.2007) = 0$$

$$\therefore \lambda = 12.2007, \lambda = 0$$

find Eigen vector $(S_w^{-1} S_B - \lambda I) \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = 0$

$$w_1 = \begin{pmatrix} -0.5755 \\ 0.8178 \end{pmatrix}, w_2 = \begin{pmatrix} 0.9088 \\ 0.4173 \end{pmatrix} = w^*$$

$w^* = S_w^{-1} (\mu_1 - \mu_2) \Rightarrow$ direct method to calculate w_2

Now calculate linear discriminant by taking the dot product of eigen vectors and original data

x_1	4	2	2	3	4	9	6	9	8	10
x_2	2	4	3	6	4	10	8	5	7	8
1 st LD	4.46	3.48	3.06	5.2	5.3	12.35	8.8	10.2	10.19	12.42

Plot LD

